World Development 140 (2021) 105332

Contents lists available at ScienceDirect

World Development

journal homepage: www.elsevier.com/locate/worlddev

Impacts of improved biomass cookstoves on child and adult health: Experimental evidence from rural Ethiopia

Daniel LaFave [a,*], Abebe Damte Beyene [b], Randall Bluffstone [c], Sahan T.M. Dissanayake [c], Zenebe Gebreegziabher [d], Alemu Mekonnen [e], Michael Toman [f]

[a] Colby College, Waterville, ME, USA
[b] Environmental Economics Policy Forum for Ethiopia, Addis Ababa, Ethiopia
[c] Portland State University, Portland, OR, USA
[d] Environmental Economics Policy Forum for Ethiopia and Mekelle University, Addis Ababa, Ethiopia
[e] Environmental Economics Policy Forum for Ethiopia and Addis Ababa University, Addis Ababa, Ethiopia
[f] The World Bank, Washington DC, USA

ARTICLE INFO

Article history:

JEL Codes:
I15
O13
Q55
Q23
Q42

Keywords:
Height-for-age
Child growth
Household air pollution
Cookstoves
Ethiopia

ABSTRACT

We present the three-year impacts of an improved biomass cookstove on child and adult health in rural Ethiopia. The *Mirt* stove is designed to cook *injera*, the staple bread of Ethiopia estimated to consume 60 percent of household cooking energy and about half of all energy in the country. Though it would not be considered a fully "clean" cooking technology and is not used on a daily basis, *Mirt's* documented fuelwood reduction potential and application to the highly energy-intensive baking of *injera* raises the question of whether it might offer health benefits to users. After near complete stove adoption during an initial one-year randomized controlled trial, 60 percent of treatment households continued to use the improved stoves three-years on and experience moderate reductions in hazardous airborne particulate matter. In a pre-specified analysis, we find treatment status is associated with a precisely estimated 0.3–0.4 standard deviation improvement in height-for-age of young children exposed to the stoves during their first years of life–a substantial effect with implications for greater health and well-being throughout the life course. This association notwithstanding, we find no changes in the respiratory symptoms or physical functioning of older children and adult cooks. Measures of fine particulate matter taken within study households remain over an order of magnitude higher than WHO standards, but follow a statistically significant gradient with respect to the observed height-for-age improvements. The results advance understanding of the health impacts of hazardous air pollution while also refining design and implementation options for interventions geared toward improving well-being in similar environments.
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1. Introduction

The disease burden from indoor air pollution results in an estimated 4 million premature deaths per year and is one of the largest drivers of poor health across the world (Lim et al., 2010). While the diminished quality of air in outdoor environments is, appropriately, a growing concern, the burning of solid biomass materials for cooking and heating *within* homes is the primary source of the pollution burden (Smith et al., 2012). Forty percent of the world relies on solid fuels such as wood, coal, charcoal, and dung for household cooking, with resulting emissions disproportionately harming women and young children in the form of lower-respiratory infections, pneumonia, low birth weights, growth deficits, and fatigue (Alexander et al., 2018; Dadras & Chapman, 2017; Fullerton et al., 2008; Pope et al., 2010; Walker et al., 2007; WHO, 2016).

In addressing these and other health concerns, organizations have sought to improve the quality of indoor air by reducing indoor burning of biomass fuels. The World Health Organization (WHO) identifies a number of clean cooking technologies and fuels, such as electricity, biogas and natural gas, that are expected to meet WHO household air pollution guideline values including a 24-hour average limit of fine particulate matter of 25 μg/m^3 (WHO,

* Corresponding author.
E-mail address: drlafave@colby.edu (D. LaFave).

https://doi.org/10.1016/j.worlddev.2020.105332
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2006a, 2015). However, these technologies and fuels are often cost prohibitive or unavailable in low and lower-middle income countries where the number of households reliant on biomass fuels is estimated to grow for the foreseeable future (IEA, 2017).

The WHO also recognizes the potential contributions improved biomass cookstoves can make to reducing household air pollution, but cautions that many stoves, including the stove considered here, do not meet their set guidelines (WHO, 2011, 2019). Improved cookstove technologies (ICTs), which increase efficiency and reduce emissions by more thoroughly burning biomass fuels, are commonly recommended as an intermediate way for households to transition away from open-fire cooking (Jeuland and Pattanayak, 2012) and improve health outcomes if demonstrably clean (WHO, 2016). Due to the potential benefits of ICTs in Ethiopia, the setting of this study, the Federal Government has proposed distributing ICTs to 30 million households by 2030 and identified reducing demand for biomass by increasing its fuel efficiency as a strategic priority in the energy sector (FDRE, 2011, 2015).

This paper contributes to a growing body of work that assesses health and environmental benefits of improved biomass cookstoves in field settings (Bensch & Peters, 2015; Hanna et al., 2016; Mortimer et al., 2017; Smith et al., 2007; Quansah et al., 2017). We examine the health impacts of the *Mirt* stove, a particular type of ICT engineered to bake *injera,* the Ethiopian staple bread. *Injera* is typically cooked 2–3 times per week in rural areas, or approximately 14 percent of a household's weekly cooking events. Estimates suggest only 2–6 percent of households use ICTs in rural Ethiopia (Central Statistics Agency of Ethiopia, 2013; Mondal et al., 2018), with most relying on tripods made from three large stones. This flexible, but inefficient, technology wastes approximately 90 percent of fuelwood energy input (Alem et al., 2014).

Working with a representative set of approximately 500 households across 36 rural communities, we disseminated *Mirt* stoves to a random subset of households and study stove use, indoor air quality, and health outcomes of women and children three and a half years after the initial randomization. While tailored to a specific food, and not intended to entirely replace all traditional cooking methods but instead be used two or three times per week, the *Mirt* stove is well suited to the context and research question as firewood for baking *injera* constitutes about 60 percent of household fuelwood consumption (Gebreegziabher et al., 2012) and approximately half the primary energy in Ethiopia (Bizzarri, 2010; Tesfay et al., 2014).

Although not a "clean" gas or electric stove (Jetter et al., 2012), *Mirt* has been shown to be significantly more efficient than traditional cooking methods. Field-based controlled cooking tests found 20–30 percent reductions in fuelwood use (Gebreegziabher et al., 2018) and survey evidence finds 40–50 percent reductions (Dresen et al., 2014; Megen Power, 2008). Evidence from laboratory-based controlled cooking tests suggests *Mirt* reduces carbon monoxide emissions by 92 percent and fine airborne particulate matter by 30 percent relative to traditional open-fire cooking (GIZ, 2007). Given what is known about the shape of the pollution-health relationship (e.g. Burnett et al., 2014; WHO, 2016) and the likely significant share of household emissions due to energy intensive *injera* cooking, the *Mirt*'s estimated efficiency improvements could potentially lead to important health benefits even if not used on a daily basis.

We build on an initial one-year impact evaluation of *Mirt* dissemination that showed users highly valued the benefits provided by the stoves and that they saved wood when cooking standardized *injera* batches (Gebreegziabher et al., 2018). We also found the *Mirt* stoves were adopted and used by over 90 percent of treatment households in the 12 months after their distribution (Beyene et al., 2015). This rate is significantly higher than a number of high-profile prior trials involving improved stoves (e.g. Hanna et al., 2016; Mobarak et al., 2012; WHO, 2006a), and in line with recent work studying low-cost stoves promoted in Africa that appear to be well-suited to the local environment and cooking practices (e.g. Bensch & Peters, 2015; Rosa et al., 2014).[1]

In the present study, we focus on health outcomes that may benefit from several years of reduced indoor air pollution. Following a pre-specified analysis plan that was written prior to data collection and then uploaded to the American Economic Association RCT registry as part of this journal's review process (Bluffstone et al., 2019a), we estimate the impact of the *Mirt* stove on child growth, a marker of early-life inflammation and health that is sensitive during the first years of life and a powerful predictor of wellbeing in adulthood (Hoddinott et al., 2013). We complement these estimates with those for symptoms of respiratory disease and measures of physical function for older children and adult cooks.[2] Comparing those randomly offered the *Mirt* stove with those that were not, our findings suggest the *Mirt* treatment increased height-for-age for children exposed to the technology before the age of three by approximately 0.3–0.4 standard deviations, while the respiratory symptoms and physical functioning of adult cooks and older children did not improve. These observed patterns are consistent with a critical early period in life where accumulated exposure may cause lasting damage and where health interventions often see their largest positive impacts (e.g. Cusick & Georgieff, 2016; Walker et al., 2007). The comparison across treatment and control households is supported by within household estimates that focus on those exposed during the sensitive early-life period to isolate the *Mirt* treatment effect from confounding factors.

The observed effect on child growth is both precisely estimated and meaningful in magnitude. A wide body of evidence links early-life growth to adult health, educational attainment, cognitive function, and earnings (e.g. Adair et al., 2013; Almond et al., 2018; Fogel, 2004; Glewwe & Miguel, 2008; Grimard & Laszlo, 2014; Groppo & Kraehnert, 2016; Hoddinott et al., 2008; LaFave & Thomas, 2017; Strauss & Thomas, 1998; Wisniewski, 2010). For example, analysis of the Ethiopian Rural Household Survey in Dercon and Porter (2014) suggests the observed height gains in the treatment group would result in approximately 3–4 percent increases in annual earnings as an adult.[3]

To better understand one of the potential mechanisms behind the effects, we simultaneously collected measures of fine airborne particulate matter with a diameter of 2.5 μm or less ($PM_{2.5}$) in a random subsample of study households. These concentrations average approximately 1297 $\mu g/m^3$ for 24-hour exposures in the control group, or about 50 times more than the WHO's threshold for healthy exposure (WHO, 2006b, 2015), and decreased by a statistically insignificant 10 percent on average for households in the

[1] As Bensch and Peters (2015) note, the complexity of different stoves and their similarity with existing cooking technologies are critical factors in stove adoption. Jetter et al. (2012) provide estimates of emissions and efficiency for 22 different cookstoves, but do not evaluate the *Mirt* ICT. While we focus here on health outcomes from an improved biomass stove, we refer the reader to Bensch and Peters (2015), Bensch and Peters (2018), Bensch et al. (2015), Beyene et al. (2015), Cundale et al. (2017), Langbein et al. (2017), Levine et al. (2018), and Lewis and Pattanayak (2012) for detailed cookstove adoption studies.

[2] While child growth is also used as a marker of nutrition, a significant and growing body of work notes physiological links between household air pollution from biomass cookstoves and reduced height-for-age (e.g. Balietti and Datta, 2018; Edwards and Langpap, 2012; Ghosh et al., 2011; Kim et al., 2016; Kyu et al., 2009; Mishra and Retherford, 2007; Pongou et al., 2006; Tielsch et al., 2009). A related literature links early life exposure to ambient air pollution and child height (Goyal and Canning, 2018; Spears et al., 2019; Tan-Soo and Pattanayak, 2019).

[3] Dercon and Porter (2014) estimate a 1cm increase in adult height results in a 1 percent increase in earnings. Our observed 0.3 to 0.4 standard deviation impact is approximately a 2cm increase in height at age 3, or 3 to 4cm as an adult.

Mirt treatment arm.[4] Households with young children, which also cook more frequently, experienced reductions of 24 percent on average (although again imprecisely estimated). When examining whether measured health outcomes correlate directly to particulate matter, we demonstrate a statistically significant gradient between reduced household air pollution and increased child growth, despite the high level of average $PM_{2.5}$, and do not find a precisely estimated or otherwise meaningful relationship between the level of particulate matter and respiratory symptoms or adult physical functioning. This evidence contributes to understanding the critically important relationship between pollution exposure and health at high levels of particulate matter (Martin et al., 2013).

The results of this study advance our understanding of the health impacts of hazardous air pollution while also refining design and implementation options for interventions geared toward mitigation in similarly situated environments. We contribute to a growing effort to study the adoption, emissions, and health benefits of improved cookstoves in multi-year field settings. Results from prior studies in this literature have often been mixed, with Hanna et al. (2016) finding low levels of adoption with no related respiratory health benefits at the end of a four-year randomized controlled trial in India, and Mortimer et al. (2017) observing no statistically significant impacts of an improved stove intervention on the incidence of pneumonia among children in Malawi. In contrast, work by Bensch and Peters (2015) finds positive health benefits on self-reported symptoms from improved stoves in a year-long RCT of 250 households in rural Senegal, which is consistent with observational studies in, for example, Cameroon (Pongou et al., 2006), Guatemala (Edwards & Langpap, 2012), India (Balietti & Datta, 2018; Mishra & Retherford, 2007), and Indonesia (Kim et al., 2016).

While health benefits for adults and older children do not arise in our data, our results suggest mitigation of indoor air pollution to be one of many important ways to improve child health and well-being. Given the link between early-life health and later life educational attainment, cognition, health, and earnings, the demonstrated benefits for young children represents a substantial and meaningful impact.

2. Study design

2.1. Ethiopian context and Mirt improved cookstove technology

Despite enjoying economic growth over the preceding decade, Ethiopia is known as a "hotspot" for forest depletion, and is one of the top four countries in the world to simultaneously maintain a high level of nonrenewable fuelwood consumption per capita and disease burden from household air pollution (Bailis et al., 2015). Neonatal disorders and lower respiratory infections, both linked to household air pollution (Amegah et al., 2014; Smith et al., 2000; Pérez-Padilla et al., 2010), are two of the top three leading causes of death (GBD, 2018), and improved biomass cooking technology use is virtually nonexistent in rural areas (Mondal et al., 2018).

This study focuses on the dissemination and use of *Mirt* stoves in rural Ethiopia. *Mirt* (translated as "best") stoves are a locally developed technology specifically designed to cook *injera,* the staple bread of Ethiopia. The stove was designed by the Ethiopian Ministry of Water and Energy in collaboration with the German aid agency GIZ, is produced from locally available raw materials, and sells at a market price of 100–250 Birr (approximately $10.00 USD at the start of the study). Repeated laboratory cooking trials conducted in conjunction with developing the stove suggested the *Mirt* reduced fuelwood use by 50.7 percent, carbon monoxide emissions by 92.3 percent, and particulate matter by 31 percent compared to traditional open-fire or three-stone technologies (GIZ, 2007).[5] Given the estimated fuel savings and a typical *injera* baking frequency of two to three times per week, the financial break-even period of two months compares favorably with the stove's five-year life expectancy.

Though the *Mirt* stove has been promoted for many years by GIZ and the Federal Government, very few households in rural settings have access to the technology or other improved cookstoves. No improved stoves were present in our study households at baseline, and data from the 2013 Ethiopian Rural Socioeconomic Survey indicate that 98.4 percent of households in rural areas cooked exclusively over traditional open-fires (Central Statistics Agency of Ethiopia, 2013).

2.2. Initial randomized controlled trial and adoption estimates

We conducted an initial one-year randomized evaluation of *Mirt* stove adoption, its fuel savings, and users' willingness to pay for the product in 2013 (Gebreegziabher et al., 2018). The baseline of the study surveyed households from 36 communities (*Gots* or sub-*Kebeles*) randomly selected to represent the forest cover in Amhara, Oromiya, and Southern Nations, Nationalities, and Peoples' (SNNP) regional states. The three states cover approximately 70 percent of the land area and 80 percent of the population of Ethiopia.

Within each enumeration area, we randomly drew 14 households from a local census and randomly assigned them into treatment and control groups at the household level within each community. The ten households in the treatment group received a *Mirt* stove and were allocated an additional behavioral treatment based on price (free vs. subsidized), incentives based on use (none vs. a small payment for recorded use), or a social networking intervention (home training only vs. home training plus community training). The control and treatment households were balanced on covariates at baseline, as well as across arms within the treatment group, and there were no other health-related behavioral change communication programs being implemented in enumeration areas throughout the study (see Beyene et al., 2015 for detail on the sampling and baseline analysis).[6]

Stove adoption choices were assessed with both self-reports by the primary cook within the household and by electronic stove-use monitors (SUMs) attached to each stove to record time and temperature readings. Enumerators returned to the households to record SUM measurements four times throughout the first year following the initial stove distribution. Adoption was high and remained high throughout the year; 12 months after the intervention, use rates were approximately 90 percent with no statistically significant differences across the treatment arms. The frequency of use remained high as well, with treatment households averaging 2.5 cooking events per week on the *Mirt* stoves, a rate consistent with the traditional frequency for baking and storing *injera* (Beyene et al., 2015).

Important for interpretation of the health results presented below, results from the 12-month follow-up suggest virtually all households chose to continue using traditional three-stone open-

[4] Similar levels of $PM_{2.5}$ have been observed in traditional open-fire cooking environments in India (Dutta et al., 2007) and Nepal (Chen et al., 2016).

[5] While based on a small sample of tests, both the reported carbon monoxide and particulate matter reductions from the use of the *Mirt* stove in GIZ (2007) are statistically significant (p-value < 0.001 for CO and p-value=0.019 for $PM_{2.5}$).

[6] Balance implies baseline demographics are unrelated to treatment status. At the household level, a regression of *Mirt* treatment status on baseline characteristics returns a p-value of 0.19 for the joint overall significance of household factors. A similar test at the individual level including age, gender, and parental characteristics returns a p-value of 0.35.

fire stoves for non-*injera* cooking. *Mirt* stoves are highly specialized for cooking *injera*, with the main burner approximately 50 cm in diameter, leaving it necessary to have a second stove to cook other foods such as stews and coffee, unless *injera* is simultaneously cooked. The *Mirt* technology was adopted for energy intensive *injera* baking, but with any pollution and health improvements constrained by the continued use of open fires by both treatment and control households.

2.3. Three-year follow up and health outcomes

This study reports on the three-year follow-up investigation conducted in late 2016, approximately 40 months after the establishment of the baseline and the initial distribution of the stoves. Given the initial high adoption rates, we added collections of household indoor air pollution and health markers in the follow-up survey to assess outcomes potentially associated with adoption of the improved biomass stoves. We focused on children and adult cooks–the two groups thought to be most at risk from exposure to indoor air pollution from biomass cooking.

Households in the follow-up survey were re-interviewed with minimal and balanced attrition across treatment and control arms. Of the original 504 households, 480 were re-interviewed, yielding a 95 percent re-contact rate.[7] Uptake of the original stoves remained quite high as 60 percent of the treatment households continued to use their original *Mirt* stoves with no observed differences across the price, incentive, or networking groups (p-value = 0.67).

In the current study, we focus on three sets of health outcomes named in our analysis plan that are linked to indoor air pollution and, potentially, *Mirt* stove use: child growth, symptom reports of respiratory-related conditions, and physical function captured through activities of daily living.

With regard to child growth, young children during the key early-life developmental period are thought to be the most at risk from household air pollution as they inhale and absorb higher levels of particles given the same exposure as adults (Bruce et al., 2013; Kurt et al., 2016; Sturm, 2012). To assess health in this population, a trained enumerator measured height (or recumbent length for children under two years of age) which was then standardized into a z-score relative to a representative, well-nourished child of the same age in months and gender using the Centers for Disease Control and Prevention growth tables (CDC, 2012). Height, conditional on age and gender, is both straightforward to measure and a well-established indicator of health status during very early childhood, reflecting both the genetic endowment and the influence of the disease environment during the in utero period and the first two to three years of life–commonly referred to as the "1st 1,000 days" (Cusick & Georgieff, 2016; Martorell & Habicht, 1986; Victora et al., 2010; Waterlow et al., 1977). Child height is a powerful predictor of attained height as an adult and is thereby associated with reduced mortality and morbidity, as well as greater economic prosperity, educational attainment, and cognitive function (Adair et al., 2013; Fogel, 2004; Glewwe & Miguel, 2008; Hoddinott et al., 2008; LaFave & Thomas, 2017; Strauss & Thomas, 1998).

Understanding the biological mechanisms linking biomass smoke exposure and child growth is an active area of research (e.g. Burnett et al., 2014; Gordon et al., 2014; Rylance et al., 2015), with a recent National Institutes of Health review identifying the link between household air pollution and child growth as a key priority (Martin et al., 2013). Exposure to particulate matter from biomass smoke is a risk factor through in utero exposure as well as direct inhalation during early life (Jayachandran, 2009; Pope et al., 2010). Prenatal impacts occur both as a result of the effect of pollution on the mother and the transfer across the placenta of toxins present in wood smoke that reduce nutrient flows and disrupt the central nervous system of the fetus (Perera et al., 1998, 1999; Poursafa and Kelishadi, 2011). Additional medical evidence links exposure to wood smoke among children to increased inflammation and a weakened immune system (Rylance et al., 2015).

Second, we assess the presence of symptoms indicating respiratory disease for all children and each adult cook in the household. Participants were asked whether they've experienced a series of symptoms over the prior four weeks, including various types of coughs, difficulty breathing, wheezing, and eye problems. Prior studies show that such symptom assessments are closely linked to clinical markers of respiratory infection and chronic obstructive pulmonary disease (e.g. Bensch & Peters, 2015; Hanna et al., 2016; Pattanayak & Pfaff, 2009) and sensitive to particulate matter exposure (e.g. Duflo et al., 2008).

Finally, each primary cook within the household provided information on their physical functioning through their ability to perform regular tasks. For each activity of daily living (ADL), the cook reported whether they could do the activity easily, if they could do it by themselves but with some difficulty, whether they needed assistance, or if they could not perform the task at all. We included difficult activities, such as walking 5 km or carrying a heavy load, tasks known as intermediate ADLs, as well as basic ADLs such as standing from sitting and performing routine housework.

The symptom and ADL modules each consist of a series of questions designed to assess an underlying outcome–respiratory health and physical function, respectively. To guard against the possibility of false positives within these groups of outcomes, we adopted the recommendation of Glennerster and Takavarasha (2013) in our analysis plan to aggregate the multiple questions into measures of the total count of symptoms and limitations and indicators noting any symptom or limitation report. Our analysis plan prespecified child height-for-age as the primary, and separate, outcome of interest due to the distinct underlying physiology of child growth implying strict age cut-offs for treatment effects and the objective nature of the height data compared to the repeated self-reports for symptoms and activities of daily living. This decision followed the practice adopted in the literature to group together variables that originate from similar questions and measure a common construct into distinct families of outcomes. We also discuss additional multiple hypothesis testing adjustments that account for the potential for false positive findings across child growth, respiratory symptoms, and physical functioning in Section 5 and Appendix A.

Measurements of household air pollution (HAP) in the cooking areas during the follow-up wave were collected in a random subsample of 202 of the 480 study households. Particulate matter with an aerodynamic diameter of 2.5 μm or less ($PM_{2.5}$) was recorded using light-scattering sensors and gravimetric pumps and filters.[8] The $PM_{2.5}$ measurements allow us to explore one potential mechanism linking the *Mirt* treatment to health, but do not rule out the role other factors, such as carbon monoxide emissions, may play in the results.

[7] Supplementary Appendix Table S1 reports tests of differential attrition across treatment and control groups at both the household and individual levels, and shows that attrition is orthogonal to treatment status as well as a baseline wealth index.

[8] Household air pollution monitoring had a target sample of 216 households, 6 from each community, in order to be able to detect a 30 percent reduction in HAP across treatment and control households following Edwards et al. (2007). We collected valid data from 93.5 percent of the target households, with missing values due primarily to equipment failures and orthogonal to treatment status. Enumerators followed strict protocols developed by Berkeley Air Monitoring Group on the placement, maintenance, and timing of the equipment–these protocols are included in the Supplemental Appendix.

The indoor air pollution and health measurements were added to the follow-up wave after considering power calculations based on the baseline sample size. In the empirical analysis that follows, we pay particular attention to this fact and provide evidence that the research design does not bias the identification of our causal treatment estimates given the randomized nature of the study.

3. Empirical approach

Our baseline analysis compares outcomes across treatment and control households, with the control group representing the counterfactual if the treatment group had not been offered *Mirt* stoves. We complement this difference approach when examining child growth by exploiting the physiology of child development where height is only sensitive to interventions in the first three years of life (Cusick & Georgieff, 2016; Martorell & Habicht, 1986). This extension allows us to exploit variation between younger and older siblings to identify the effect by a difference-in-difference between children within a household. All regression approaches estimating causal effects of the treatment were pre-specified prior to data collection and stated in our analysis plan.

3.1. Baseline treatment and control comparison

Our baseline models estimate the difference across individuals in treatment and control households. For a given individual i in household h and community c, we estimate the following model:

$$y_{ihc} = \beta_0 + \beta_1 I(Treatment)_h + \delta X_{ih} + \mu_c + \varepsilon_{ihc} \quad (1)$$

where y is a health outcome and β_1 is the coefficient of interest that captures the causal effect of one's household being assigned to the treatment group. As treatment was randomly assigned and orthogonal to the vector of additional control variables measured at baseline, X, the inclusion of controls aids only in improving the precision of the estimates. Additional covariates include the following: gender; flexible polynomials in years of age for adults or age in months for children; household size and demographic composition; the age, education, and gender of the head-of-household; dwelling characteristics; and a composite wealth index. Models also include community fixed effects, μ_c, to capture observed and unobserved features of local areas that may impact health such as access to services and local markets.

We observe a median of 3 children under the age of 15 per study household and are therefore able to estimate household random effects extensions of Eq. (1) for child outcomes. This approach allows the unobserved error to contain a fixed component common to all household members yet uncorrelated with treatment status– a plausible specification given the randomized nature of the treatment.

Eq. (1) estimates intent-to-treat (ITT) effects identified by the random assignment of treatment status. This causal estimate includes both the 60 percent who continued using the *Mirt* stove and the 40 percent who had stopped using it at some point during the prior three and a half years. As stoppage is potentially a confounding choice, we also present estimates of actual stove use with an instrumental variable strategy. These two-stage models replace *I(Treatment)* with a variable measuring the fraction of time since the baseline the household has used the stove (Hanna et al., 2016). This variable takes a value of 1 for households still using the stove. For those who are not using the *Mirt* at endline, use is measured based on prior available SUM temperature sensor readings and reports by the cook on when the household stopped using the stove. The mean of this stove use measure is 0.89, suggesting that the 40 percent of households who had stopped using the stove at the time of the follow-up survey used the *Mirt* for approximately 29 of the 40 months on average. Randomly assigned treatment status then serves as an instrumental variable for stove use in a first stage. As some from the treatment group do not use the *Mirt* stove but households in the control group were not observed to start using *Mirt* on their own, our setting satisfies the necessary condition of one-sided noncompliance to interpret this estimate as the average treatment effect of actual *Mirt* use (Imbens & Rubin, 2015).

3.2. Child growth

Our analysis of child linear growth, also referred to as height-for-age, makes use of the additional feature that the timing of the intervention within a child's life is critical. A significant literature in nutrition and epidemiology defines the first 1,000 days of a child's life to be a crucial period when health interventions can significantly affect child height (e.g. Currie & Vogl, 2013; Hoddinott et al., 2008; Martorell & Habicht, 1986; Victora et al., 2010), after which reduced exposure to indoor air pollution is expected to have no effect on growth (e.g. Berkey et al., 1984; Ghosh et al., 2011; Goldizen et al., 2016; Kyu et al., 2009; Tang et al., 2006; Sram et al., 2013). Given the significance of this cut-off, improvements in height-for-age should only occur for those children young enough to benefit from the *Mirt* treatment. Similar identification strategies based on the biology of height-for-age have been used to examine the child health impacts of primary care services (Frankenberg et al., 2005), pension income (Duflo, 2003), cash transfers (Farfan et al., 2012), and supplementary child feeding programs (Giles & Satriawan, 2015), among others.

With forty months between the baseline and follow-up waves, we're able to separate children into two groups based on their age and expected effect of exposure to the *Mirt* stove: (Fig. 1)

The fact that only very young children in treatment households should benefit from the intervention allows for within-household comparisons to estimate the treatment effect; conditional on age and gender, young children in treatment households should be closer in height to their older siblings than young children in control households are to their older siblings.

This intuition maps to a regression model with the following modification of (1):

$$z_{ihc} = \gamma_1 I(Treat\ \&\ BaseAge < 36mo.)_{ih} + \gamma_2 I(Treat\ \&\ BaseAge \geq 36mo.)_{ih} + \delta X_{ih} + \mu_h + \varepsilon_{ihc}$$

where a child's height-for-age z-score is the dependent variable. Additional baseline controls in X include the same factors as in model (1) as well as controls for parental height to capture the

Age at baseline (2013)	Age at follow-up (2016)	Expected Effect: Treatment	Expected Effect: Control
< 36 months (or not born)	< 76 months	(+)	0
≥ 36 months	≥ 76 months	0	0

Fig. 1. Expected child height-for-age effects by treatment status and age.

Table 1
Descriptive Statistics.

Means and Control vs. Treatment Differences				
	All (1)	Control (C) (2)	Treatment (T) (3)	C vs. T (p-value) (4)
A. Child Outcomes (n=1152)				
Age	8.378	8.359	8.386	0.920
	(0.121)	(0.231)	(0.142)	
Female	0.492	0.452	0.507	0.078
	(0.014)	(0.026)	(0.017)	
Child Growth				
Height-for-age (z-score)	−1.220	−1.263	−1.203	0.532
	(0.043)	(0.082)	(0.050)	
Symptom Reports				
Any reports	0.463	0.516	0.443	0.021
	(0.014)	(0.027)	(0.017)	
Any cough reports	0.242	0.271	0.231	0.137
	(0.012)	(0.024)	(0.014)	
N. symptoms reported (max: 20)	2.041	2.262	1.954	0.142
	(0.094)	(0.183)	(0.110)	
B. Adult Cook Outcomes (n=475)				
Age	37.609	37.557	37.571	0.963
	(0.975)	(0.601)	(0.511)	
Female	0.922	0.915	0.917	0.763
	(0.021)	(0.013)	(0.011)	
Activities of Daily Living (ADL) Reports				
N. ADLs with any difficulty	3.896	3.947	3.933	0.867
(max: 12)	(0.262)	(0.161)	(0.137)	
Symptom Reports				
Any symptom reports	0.787	0.773	0.777	0.720
	(0.032)	(0.020)	(0.017)	
Any cough reports	0.329	0.377	0.364	0.278
	(0.037)	(0.024)	(0.020)	
N. symptoms reported (max: 20)	4.726	4.677	4.690	0.911
	(0.375)	(0.227)	(0.194)	
C. Household Level (n=475)				
Household size	5.975	5.870	6.018	0.462
	(0.091)	(0.157)	(0.111)	
N. Children (ages 0–14)	2.521	2.399	2.570	0.284
	(0.072)	(0.132)	(0.087)	
N. Prime age (ages 15–59)	3.181	3.159	3.190	0.833
	(0.066)	(0.124)	(0.078)	
Age of household head	46.419	46.993	46.187	0.524
	(0.571)	(1.108)	(0.666)	
Female headed household	0.083	0.051	0.096	0.101
	(0.013)	(0.019)	(0.016)	
Stove use (share of months since	0.632	0.000	0.888	0.000
baseline using improved stove)	(0.020)		(0.011)	
Indoor Air Quality				
log 24-hr Mean $PM_{2.5}$	6.341	6.414	6.263	0.428
	(0.095)	(0.134)	(0.135)	
log 24-hr Maximum $PM_{2.5}$	10.463	10.469	10.455	0.909
	(0.060)	(0.083)	(0.089)	

Notes: Table reports means and standard errors of key variables for the entire sample as well as separately for control and treatment groups. Column 4 reports p-values of a means test across the control and treatment groups. Panels A and B report measures at the individual child and cook level. Panel C reports measures at the household level.

genetic determinants of height. The parameter of interest, γ_1, measures the relative change in height for children in the treatment group who were under 36 months old or not yet born at the time of the baseline.[9]

Compared to the treatment versus control difference in Eq. (1), the model in Eq. (2) incorporates two key additional features. First, the inclusion of a household fixed effects, μ_h, is now possible, as treatment effectively varies at the individual level. This allows for the fixed effect to capture any shared characteristics common to all children in a household and restricts identification to within-household comparisons. This strategy accounts for a large set of potentially confounding variables that have plagued past associations of stove use and health. Second, it identifies an expected placebo effect for those treatment children three years and older in the baseline wave: compared to children in the control group of the same age, these children should see no change in height and γ_2 should be zero.

4. Summary statistics and descriptive evidence

Table 1 reports descriptive means and standard errors of key variables of interest. Panels A and B report measures at the individual level for children under the age of 15 and adult cooks. Columns

[9] This regression specification was pre-specified in our analysis plan. The validity of this approach requires that fertility did not respond to the intervention in a way that might bias the sample of new, young children in the 2016 follow-up. Supplementary Appendix Table S2 reports results from regressing the number of new children in a household against treatment status, showing the two have no relationship.

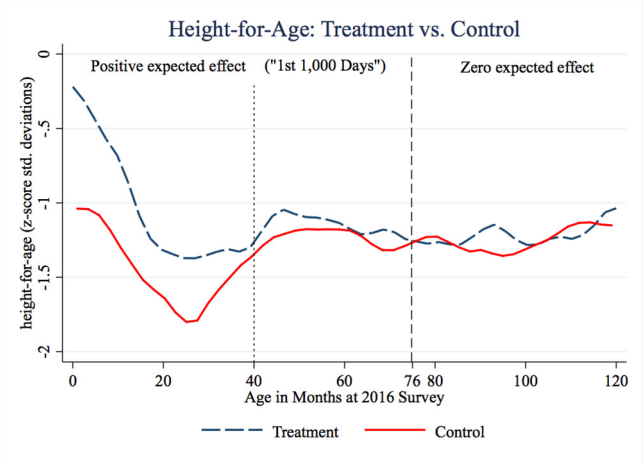


Fig. 2. Treatment and Control Height-for-Age Comparison.

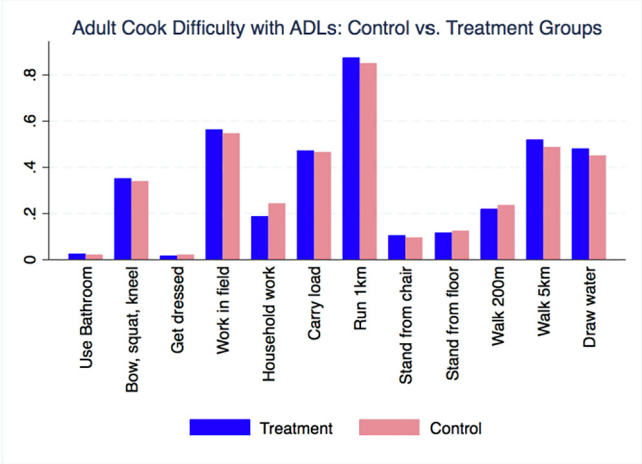


Fig. 4. Mean of adult cook ADLs across treatment and control.

2 and 3 distinguish between the control and treatment households, and Column 4 reports *p*-values for tests of equality across the two groups.

Mean height-for-age in the sample is −1.2 standard deviations, suggesting that the average child is 1.2 standard deviations shorter than a healthy child of the same age (in months) and sex. Comparing across treatment and control suggests that the treatment group is 0.06 standard deviations taller, but the difference is not statistically significant, as seen in column 4.

Fig. 2 illustrates the treatment and control height-for-age measure for children up to age 10, constructed by nonparametric regressions of height-for-age on age in months at the 2016 follow-up survey. We illustrate the sensitive period of growth with vertical lines as those children exposed during their first three years of life were up to 76 months old in the follow-up survey.

The overall mean level and general trend with respect to age are consistent with commonly used nationally representative datasets, including the 2011 Ethiopian Demographic and Health Survey (DHS) and Living Standards Measurement Surveys (LSMS), and seen across a variety of developing settings in population-representative data (e.g. Strauss & Thomas, 2007).

Fig. 2 shows that there is a marked difference in the height of children at early ages when comparing across treatment and control households. As the intervention had been in place for 40 months, children in the treatment group ages 40 months and younger have had potentially complete exposure to the *Mirt* stove, while those between 40 and 76 months have an amount of exposure decreasing in their age. Regression models below further examine these differential effects.

Fig. 3 represents the corresponding treatment and control comparison for child and adult symptom reports. Children report an average of 2.3 symptoms in the control group compared to 2.0 in the treatment group, although the difference is not statistically significant (*p*-value = 0.14). The gaps are generally quite small between the groups but tend toward the expected direction with treatment individuals 8 percentage points less likely to report any symptoms at all (*p*-value = 0.02). Regression results below assess the gap between treatment and control including additional covariates.

Fig. 4 summarizes adult cooks' reported difficulties with activities of daily living. For each question, the four-step scale is collapsed to a binary indicator equal to one if the respondent reports severe or moderate difficulty with the activity. As in Fig. 3, the differences across the groups are minimal and statistically insignificant. Lack of significant differences across the groups persists using alternative severity thresholds.

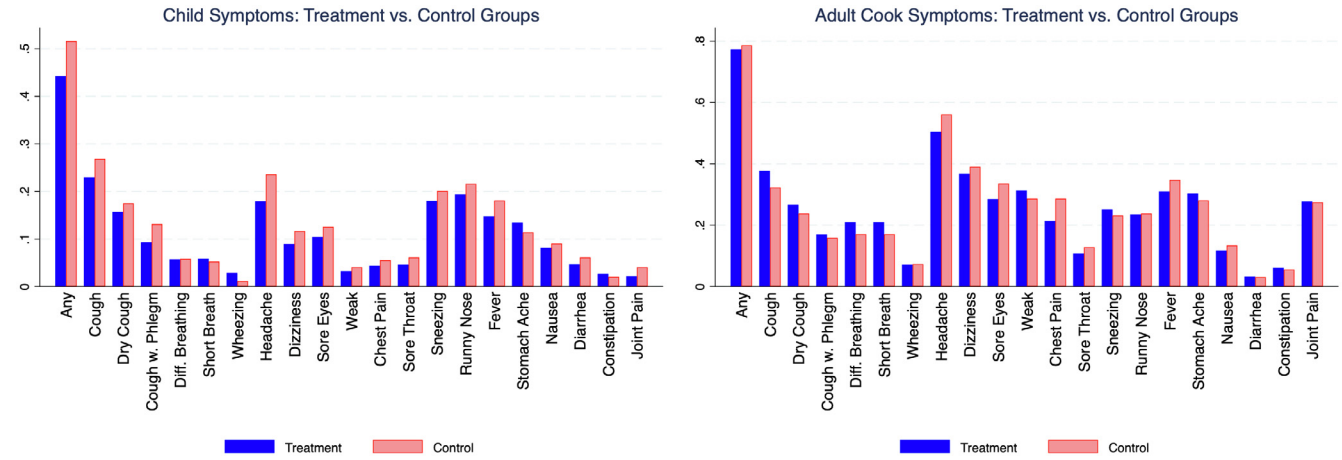


Fig. 3. Means of child and adult symptom reports across treatment and control households.

5. Treatment effect estimates

5.1. Child growth

Table 2 presents estimates of the impacts of treatment status and stove use on child growth, our primary outcome of interest, from Eq. (2). Columns 1 through 3 focus on intent to treat effects of treatment assignment. All models include additional individual and household level controls and standard errors are adjusted for clustering at the household level given the multiple children per household structure of the data.

Column 1 presents treatment estimates for both those children exposed while in the sensitive period and older children. The estimates suggest being in a household that was offered a *Mirt* stove while in the sensitive period of growth is associated with a 0.277 standard deviation increase in predicted height-for-age relative to the control children of the same age. This is a sizeable effect and similar in scale to the estimated benefits of improved non-biomass cookstoves on birth outcomes, such as gestational age and birth weight (e.g. Alexander et al., 2018), suggesting part of the mechanism may work through in-utero exposure.

The underlying physiology predicts there should be no impact on older children in the treatment group. The point estimate of −0.013 standard deviations (p-value = 0.92) supports this conclusion, as treatment children older than 36 months at the baseline saw no benefit in terms of their height-for-age compared to children in the control group. This is in-line with established literatures in nutrition and epidemiology, and supports the experimental design of our study as it implies there were no pre-existing differences in height-for-age across treatment and control children.

Column 2 of Table 2 exploits the multiple children per household structure of the data and includes household random effects. Estimates in Column 2 suggest a precisely estimated 0.367 standard deviation treatment effect for young children (p-value = 0.013) and a null effect for older children.

Column 3 is the most demanding of the specifications as it includes household fixed effects and is identified only from comparisons of siblings within the same household. Given the within-household design, it is not possible to identify the effects for both younger and older children presented in columns 1 and 2. All observed and unobserved characteristics common at the household level are absorbed into the fixed effect, and biases threatening the causal interpretation of the estimates would have to vary within treatment and control households in such a way as to be correlated both with treatment status and only positively impact the youngest children.

The 0.383 z-score estimate in Column 3 suggests that, conditional on their age and gender, young children in treatment households are closer in height to (or relatively taller than) older children in their household compared to young children in control households. The result is again precisely estimated (p-value = 0.023). This is a large and meaningful finding, corresponding to approximately a 2 cm gain for a 3-year-old child.

Columns 4 through 6 report treatment effects of stove use measured as the share of months the household used the stove. These results are instrumental variable estimates where stove use-by-age interactions are instrumented with treatment status interacted with the corresponding ages. The results suggest the use of the stove over the full period relates to gains in height-for-age between 0.32 and 0.45 standard deviations for children in the sensitive period of growth, with no benefits for older children. Both results are important—these are sizable gains for young children, and the expected placebo effect for older children again supports the validity of the empirical approach.

Although our analysis plan ranked the health outcomes in order of importance with child height-for-age listed first, one may be concerned about the statistically significant results in Table 2 appearing by chance in light of the multiple symptom and physical functions outcomes collected alongside child growth. At the suggestion of a helpful reviewer, Appendix A and Appendix Table A.1 present adjusted p-values for the height-for-age treatment effects that account for the multiple outcomes in the study. These adjustments use a modified Bonferroni correction following Sankoh et al. (1997) that accounts for the highly correlated nature of the self-reported symptom and limitation measures. The adjusted p-values and joint significance tests presented in Appendix Table A.1 strengthen the credibility of the height-for-age findings in Table 2 as the pre-specified random and fixed effects estimates remain statistically significant at either the 5 or 10 percent level when using randomization

Table 2
Effects of Improved Stoves on Child Height-for-Age.

	Dep. Var.: Height-for-age (z-score std. deviations)					
	Intent to Treat Effects (ITT)			Stove Use Effects		
Model Specification	OLS	RE	FE	IV	IV-RE	IV-FE
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment × Baseline Age < 36 mo.	0.277*	0.367**	0.383**			
	(0.153)	(0.148)	(0.168)			
Treatment × Baseline Age ≥ 36 mo.	−0.013	0.024				
	(0.140)	(0.135)				
Stove Use × Baseline Age < 36 mo.				0.320*	0.425**	0.447**
				(0.172)	(0.170)	(0.195)
Stove Use × Baseline Age ≥ 36 mo.				−0.016	0.026	
				(0.155)	(0.153)	
Additional controls	yes	yes	yes	yes	yes	yes
Household random effects		yes			yes	
Household fixed effects			yes			yes
Observations	1,152	1,152	1,152	1,152	1,152	1,152

Notes: Table reports intent to treat estimates of stove offers in columns 1–3 and instrumental variable effects of stove use in columns 4–6. Columns 2 and 5 include household level random effects and columns 3 and 6 include household fixed effects. Dependent variable is child height-for-age (z-scores), standardized according to the CDC growth tables. Sample is all children ages 15 and under. All models include a cubic polynomial in age in months, sex, mother's height and age, and father's age. Household level controls are omitted from regressions with household fixed effects, and include household size and demographic composition, age, education and sex of the household head, dwelling characteristics, community fixed effects, and a composite wealth index. Standard errors clustered at the household level in parentheses. ** p < 0.05, * p < 0.1.

Table 3
Effects of Improved Stoves on Symptoms and Physical Functioning.

	Children			Adult Cooks				
	N. Symptoms	Any Symptoms	Any Cough	N. Symptoms	Any Symptoms	Any Cough	N. ADLs with Difficulty	Unable to Do any ADLs
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Intent to Treat								
Treatment	−0.116	−0.048	−0.018	0.042	−0.014	0.051	0.192	0.023
	(0.251)	(0.037)	(0.035)	(0.414)	(0.038)	(0.050)	(0.287)	(0.043)
Use Effects								
Stove Use	−0.132	−0.054	−0.020	0.048	−0.016	0.057	0.217	0.026
	(0.278)	(0.041)	(0.039)	(0.430)	(0.039)	(0.052)	(0.299)	(0.044)
Additional Controls	yes	yes	yes	yes	yes	yes	yes	yes
Control group mean	2.262	0.516	0.271	5.243	0.846	0.353	3.868	0.507
Observations	1,147	1,147	1,147	475	475	475	475	475

Notes: Table reports estimates of the impact of stove offers and stove use. ITT and stove use estimates come from separate regressions. The dependent variable is the count of symptoms or activities in columns 1, 4, and 8 and linear probability models with binary outcome indicators for remaining columns. Sample is all children ages 15 and under in columns 1–3 and adult cooks 16 and older in columns 4–8. Additional controls include a cubic polynomial in age in months, sex, mother's age and height, father's age, household size and demographic composition, household head age, education, and sex, dwelling characteristics, community fixed effects, and a wealth index. Standard errors clustered at the household level for child outcomes and community level for adult outcomes in parentheses.

inference or adjusting for three, six, or eight additional symptom and physical functioning outcomes. Please see Appendix A for further discussion.

5.2. Respiratory symptoms and adult physical functioning

Table 3 summarizes the ITT and use impacts on acute respiratory symptoms and measures of physical functioning for older children and adults. Unlike height-for-age, these domains are potentially malleable across the age distribution, thus estimates are determined by differences across households. Models include community fixed effects and individual and household level controls as in equation (1), and standard errors are adjusted for clustering.[10] Supplementary Appendix Tables S3, S4, and S5 report results for the 20 individual symptoms and 12 activities of daily living used to construct the summary measures in Table 3 for transparency, but inferences on one specific symptom or activity is cautioned against due to multiple hypothesis testing concerns.

Columns 1 through 3 report impacts on acute symptoms for older children. The results suggest there are no precisely estimated links between the *Mirt* intervention and one's symptom burden based on a four-week recall period. There is no evidence of statistically significant intent to treat or use effects for the total number of symptoms in Column 1, an indicator for any symptoms in Column 2, or for cough specific symptoms in Column 3.[11]

While the point estimates in columns 1 through 3 are negative and would suggest reduced symptom loads for children in treatment households, the results should be interpreted as suggestive at best. For the 20 specific symptoms in Supplementary Appendix Table S3, we estimate one reduction at the five percent significance level on headaches and one effect at the ten percent level—results which are balanced out by estimated increases in wheezing and shortness of breath. Such a pattern is consistent with pure chance given the expected error rates and the number of outcomes. Adjusting statistical significance thresholds for multiple comparisons only further pushes the results toward concluding a lack of effect.[12] The null findings also hold when examining subgroups of the population based on specific 5-year age groups (ages 5 and under, 6–10, and 11–15), regions, and baseline wealth quantiles.

The remainder of Table 3 presents results for adult cooks. Columns 4 through 6 show that there are no meaningful links between a cook's reported symptoms and either assigned treatment status or stove use. Columns 7 and 8 report effects for summary measures of the activities of daily living outcomes assessing the number of reported difficulties and the likelihood of experiencing any specific difficulties. The estimated effects are small in magnitude and not statistically different from zero.[13] The null physical functioning results hold using ordered probit models to examine the gradient of difficulties for each outcome and when considering binary outcomes for severe levels of limitations rather than the severe or moderate levels used in constructing Table 3. As with children, the null findings hold after dividing the adult cooks by age, by region, and by wealth quantile.

Taken as a whole, the results suggest a nuanced pattern of significant gains in early-life growth for young children, yet no noticeable impacts for either adult cooks or older children. We next present evidence examining the link between pollution measurements and health outcomes.

6. Mechanisms–pollution concentrations and health outcomes

As a step toward explaining the positive child growth effects and absence of other health effects, we examine the effect of the treatment on indoor air quality measures and the relationship between household air pollution and key health outcomes.

6.1. Household air pollution (HAP)

We measured indoor air quality in cooking areas with Particle and Temperature Sensor Plus (PATS+) light-scattering meters in a

[10] We cluster at the village level for adults and, to account for multiple children per household, at the household level for child outcomes. The lack of statistically significant effects shown in Table 3 are consistent with those that adjust for multiple hypothesis testing.

[11] Note that the total count and indicator outcomes were pre-specified in our analysis plan. The any cough outcome was added in response to a suggestion to be comparable to analysis in Hanna et al. (2016).

[12] This conclusion follows using any multiple comparison adjustment as such procedures are concerned with false rejections of a true null and there are no statistically significant treatment effects present in Table 3 using conventional thresholds.

[13] While lack of statistical power for adult outcomes, rather than lack of a true effect, is also a possible explanation for null findings, adult symptom and activities of daily living indicators had ex-ante minimum detectable effects (MDE) of 0.13 and 0.17, respectively. The corresponding coefficient estimates in columns 5, 6, and 8 of Table 3 are small in magnitude, both in absolute terms and relative to the MDEs, suggesting a lack of actual ITT and use effects.

Table 4
Relationship between Health Outcomes and Indoor PM2.5.

A. Child Height-for-Age						
	Age <36 months Height-for-Age (z-score)			Age <76 Months Height-for-Age (z-score)		
	(1)	(2)	(3)	(4)	(5)	(6)
ln(Maximum PM2.5 level)	−0.638** (0.252)	−0.471** (0.177)	−0.569** (0.239)	−0.304** (0.123)	−0.239** (0.110)	−0.295*** (0.108)
Individual Level Controls		yes	yes		yes	yes
Household Level Controls			yes			yes
Observations	53	53	53	159	159	159

B. Respiratory Symptoms and Physical Functioning						
	Children		Adult Cooks			
	N. Symptoms	Any Symptoms	N. Symptoms	Any Symptoms	Any ADL limitations	N. ADL limitations
	(1)	(2)	(3)	(4)	(5)	(6)
ln(Maximum PM2.5 level)	−0.302 (0.246)	−0.007 (0.033)	−0.010 (0.357)	−0.004 (0.026)	−0.015 (0.041)	0.093 (0.239)
Individual Level Controls	yes	yes	yes	yes	yes	yes
Household Level Controls	yes	yes	yes	yes	yes	yes
Observations	513	513	202	202	202	202

Notes: Table reports results from regressions of health outcomes on log maximum indoor PM2.5 concentrations. Panel A examines child height for age for children under 36 months old in columns 1–3 and under 76 months old in columns 4–6. Panel B. examines respiratory symptoms for children 15 years and under in columns 1 and 2 and adult cook outcomes in columns 4 through 6. Individual level controls include a cubic polynomial in age, sex, mother's height and age, and father's age. Household level controls include household size and demographic composition, age, education and sex of the household head, dwelling characteristics, and a composite wealth index. Standard errors robust to clustering at the household level used for child outcomes and community level used for adult outcomes. *** p < 0.01, ** p < 0.05

random subset of study households: 99 treatments and 103 controls. The PATS+ devices record maximum, mean and median $PM_{2.5}$ levels and were in place in households' cooking areas for 72 h, a time sufficient to capture an *injera* baking event given the typical frequency of two to three baking sessions per week.[14]

The recorded levels of particulate matter in the treatment and control households are well above the current WHO 24-hour standard of 25 $\mu g/m^3$ (WHO, 2015) and consistent with measurements in similar environments (e.g. Chen et al., 2016; Dutta et al., 2007). Mean (median) concentrations in treatment and control households are 1162 (464) and 1297 (525) $\mu g/m^3$ for average 24-hour exposures. Maximum concentration levels are an order of magnitude higher.

While treatment households have approximately 10 percent lower levels of average $PM_{2.5}$ concentrations, the difference across treatment and control households is not statistically significant in means comparisons (p-value = 0.53) or regression specifications adjusting for household level covariates. Households with young children, which also cook more frequently, experienced reductions of 24 percent on average, albeit still imprecisely estimated (p-value = 0.50).[15]

6.2. Indoor air pollution and health

Although the reductions in $PM_{2.5}$ in *Mirt* households are imprecisely estimated, the simultaneous measurement of household air pollution and health outcomes allows us to descriptively assess the possibility of pollution-health gradients that are noted as lacking in the literature (Goldizen et al., 2016; Martin et al., 2013). For each of our health outcomes, we estimate regressions of an individual's health marker against the natural log of their household's particulate concentration. The results of these tests should be interpreted as suggestive evidence of a health-pollution correlation rather than the exact magnitude of the slope as the treatment had a small and imprecise effect on pollution and the shape of these exposure–response functions has only relatively recently begun to be investigated (Burnett et al., 2014), particularly at the observed levels of $PM_{2.5}$.

Panel A of Table 4 reports results with a child's height-for-age z-score as the dependent variable and the log of the maximum particulate matter concentrations as the key variable on the right-hand side. Following the known sensitive period of growth, in columns 1 through 3 we focus on children currently under 36 months of age who would be contemporaneously impacted by the measured pollution level.[16]

There is a clear, statistically significant link between reductions in log maximum $PM_{2.5}$ and improvements in child height-for-age. Column 3, which includes individual and household level controls, suggests that a 24 percent decrease in $PM_{2.5}$ relates to a 0.1 standard deviation increase in height for age ($0.24^*(e^{-0.569}-1)$).

Columns 4 through 6 of Panel A repeat the analysis including all children under 76 months old who had the potential to benefit from the intervention during their sensitive period of growth. This approach requires that pollution readings captured at the follow-up wave reflect exposure across the prior forty-month period. The negative and statistically significant relationship between maximum $PM_{2.5}$ and height-for-age persists, although the magnitude falls by approximately fifty percent. The 36–76-month group independently has a −0.201 estimate (p-value = 0.11), suggesting that the link between measured pollution and height-for-age is strongest for children in the sensitive growth period.

[14] Following best practices, gravimetric pump and filters were co-located for exactly 24 h in a random subsample of 50 households to calibrate the PATS+ results (Berkeley Air Monitoring Group, 2017). A discussion paper elaborating on the household air pollution data and presenting preliminary results is available as Bluffstone et al. (2019b).

[15] Ex-post heterogeneity analysis not included in our pre-specified analysis suggests the location of the *Mirt* stove may be an important factor for future interventions to consider: $PM_{2.5}$ was approximately 60 percent lower in *Mirt* households among those who cook in their main living area. This subsample (53 of the 202 households in the HAP sample), is of insufficient size to estimate ex-post heterogeneity in health effects along this dimension.

[16] Supplementary Appendix Table S6 repeats the analysis in Table 4 using the mean $PM_{2.5}$ concentration rather than the maximum.

While there is suggestive evidence of a pollution–height-for-age link at the observed levels of particulate concentration, we find no such link for respiratory symptoms or physical functioning. Panel B of Table 4 presents estimates for summary measures of child symptoms in columns 1 and 2, and adult cook outcomes in columns 3 through 6. All regressions include additional individual and household level controls.

None of the six estimates are statistically significant. The effects do not appear to be driven by small sample size concerns, as many of the coefficients are small in magnitude.

Taken together with Panel A, the null findings in Panel B suggest a potential pollution-health mechanism that would lead to growth effects for young children, even without observed health benefits for adult cooks and older children. This pattern is plausibly consistent with physiological scarring attributed to cumulative exposure to high levels of particulate matter in the past for adults and older children (Naeher et al., 2007).

7. Discussion

We examine the health impacts of transitioning to the *Mirt* improved biomass cookstove in a randomized trial conducted in rural Ethiopia. Though the *Mirt* stove is not a stove that may be expected to achieve WHO guideline levels for indoor air quality, in line with other studies (e.g. Bensch & Peters, 2015), we find potentially important health effects for part of our sample population. These findings suggest that improved biomass cookstoves that reduce fuelwood consumption may also offer health benefits. Three and a half years after the initial distribution of the stoves, households randomly assigned to receive the units continued to use the technology at a high rate, and young children in these households experienced gains in height-for-age on the order of 0.3–0.4 standard deviations. We interpret these findings as suggesting the *Mirt* stove, though not a fully clean cooking technology and not used for all cooking, may be one additional tool to improve child well-being alongside established evidence on the importance of broad nutrition and child health programs.

However, adult cooks and older children experienced no gains in measured respiratory symptoms or markers of physical function. While relatively little is known about the specific health benefits of pollution reductions at the high levels of $PM_{2.5}$ observed in the study, we find evidence suggesting the change in $PM_{2.5}$ due to the *Mirt* stove may be enough to cause improvements in height-for-age, though not enough to see noticeable differences in older individuals.

From a policy perspective, the child growth gains may appear modest given the null impact on adults and older children. Still, over the course of a child's life they represent significant benefits due to the later-life impacts of early-life health improvements. Children who are taller early in life are more likely to complete primary and secondary education, develop greater cognitive capacity, earn more as adults, and live longer, healthier lives. The results presented here suggest exposure to high levels of particulate matter from biomass smoke is a contributing factor to a significant health burden for young children, and that its mitigation, in some cases through promotion of improved cookstove technologies, may result in significant improvements in well-being around the world.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Financial support from the World Bank under contract 7178322 is gratefully acknowledged, as are comments from Erick Gong, John Maluccio, Lindsey Novak, Andrew Simons, and two anonymous referees. Tenzin Choephel, Emil Kee-Tui, Kelly Ling, Ryu Matsuura, Shafat Rahman, and Beatrice Woodruff provided excellent research assistance at various stages of the project. Dedicated in honor of Kirk R. Smith.

Appendix A:. Adjusted inferences for Height-for-age treatment effects

This appendix presents adjusted *p*-values for the height-for-age treatment effects and tests of joint statistical significance that account for the multiple health outcomes in the study. Our pre-specified analysis plan (Bluffstone et al., 2019a) ranked three sets of outcomes in order of importance to the research program: child height, self-reported symptoms, and activities of daily living. However, one may still be concerned that the statistically significant height-for-age treatment effects for young children in Table 2 arise due to statistical chance across the health outcomes of the study.

To account for this concern, Appendix Table A.1 reports randomization inference *p*-values as well as those that adjust for inference across child height-for-age and the symptom and ADL outcomes. Following Sankoh et al. (1997), we implement a modified Bonferroni correction that holds the Family Wise Error Rate constant while accounting for the mean correlation of the multiple outcomes – a mean correlation of zero reduces to the full Bonferroni adjustment and a correlation of one would require no adjustment. This is particularly important for our study where the self-reported symptom and ADL outcomes are highly correlated – for example, when we fail to reject a null treatment effect on the count of child symptoms in Table 3, we're also likely to fail to reject for the count of adult symptoms as the correlation between the two outcomes is approximately 0.6.

Columns 1 through 3 of Appendix Table A.1 report the coefficient, clustered standard error, and corresponding *p*-value for each of the intent to treat and stove use effects for young children in Table 2. Column 4 presents *p*-values calculated using randomization inference to guard against the clustered standard errors being potentially downward biased (Young, 2019). The adjusted OLS and random effects *p*-values slightly rise in Column 4 while the fixed effects *p*-values fall; no conclusions about significance at the 5 or 10 percent level change.

Columns 5 through 8 report *p*-values that adjust for different, and increasing, sets of symptom and ADL outcomes from Table 3: column 5 includes height-for-age with all other child outcomes (cols. 1–3 of Table 3) and the estimates maintain their significance at the 5 and 10 percent levels. Column 6 groups height-for-age with one outcome from each set of child symptoms, adult symptoms, and adult ADLs (cols. 1, 4, and 7 of Table 3). Here the height-for-age OLS coefficients are no longer statistically significant, while the random and fixed effects estimates remain significant at the 5 percent level. Columns 7 adds the incidence indicators for symptoms and ADLs (cols. 1–2, 4–5, 7–8 of Table 3) and column 8 adjusts for all of the symptom and ADL aggregates in Table 3.

While the OLS coefficients are no longer statistically significant at conventional levels in three of the four sets of MHT adjustments, the pre-specified and preferred random and fixed effects estimates remain significant at either the 5 or 10 percent level, even when adjusting for 6 and 8 additional outcomes.

Panel B of Appendix Table A.1 includes results from tests of significance of the height-for-age treatment effects jointly with the corresponding group of outcomes named in each of columns 5

Table A1
Multiple Hypothesis Adjusted *p*-values for Height-for-age Treatment Effects.

					Multiple hypothesis testing adjustments including height-for-age with [. . .]			
	Table 2 coefficient with reported *'s	Clustered Standard Error	Table 2 *p*-value	Randomization Inference *p*-value	3 additional outcomes (all child outcomes)	3 additional outcomes (symptoms and limitations)	6 additional outcomes (symptoms and limitation counts and incidence)	8 additional outcomes (all Table 3 outcomes)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Adjusted p-values for height-for-age estimates								
Intent to Treat Effects								
Column 1: OLS	0.277*	(0.153)	0.071	0.085	0.097	0.140	0.224	0.265
Column 2: Random Effects	0.367**	(0.148)	0.013	0.022	0.018	0.027	0.044	0.053
Column 3: Fixed Effects	0.383**	(0.168)	0.023	0.017	0.032	0.047	0.077	0.093
Stove Use Effects								
Column 4: OLS	0.320*	(0.172)	0.062	0.077	0.085	0.123	0.198	0.235
Column 5: Random Effects	0.425**	(0.170)	0.013	0.022	0.018	0.027	0.044	0.053
Column 6: Fixed Effects	0.447**	(0.195)	0.022	0.016	0.030	0.045	0.074	0.089
Panel B: Joint tests of significance across height-for-age and sets of other outcomes (p-values)								
			Intent to Treat Effects		0.043	0.069	0.081	0.083
			Stove Use Effects		0.033	0.057	0.071	0.072

Notes: Panel A reports coefficients, clustered standard errors, and corresponding p-values for child height-for-age regressions shown in Table 2 of the paper. Columns 4 through 8 offer adjusted *p*-values based on randomization inference (col. 4), and multiple hypothesis testing corrections utilizing the correlation of the outcomes in the adjustment (discussed in Sankoh et al., 1997). Colum 5 adjusts for the three other child outcomes in the paper, cols. 1–3 of Table 3, column 6 adjusts for the count of child and adult symptoms and ADLs included in the pre-specified analysis plan for the intervention (cols. 1, 4, and 7 of Table 3), column 7 adds adjustment for the indicators of any symptoms or ADLs (cols. 2, 5, and 7 in Table 3), and column 8 adjusts for all outcomes in Table 3. Panel B reports p-values from tests of overall treatment significance of height-for-age jointly with the corresponding set of outcomes in each column. Randomization inference and joint significance tests are done following Young (2019).

through 8. Following Young (2019) these are tests of overall experimental relevance that test against the null that the intervention had no impact across the outcomes for all individuals.

The rejections of the joint null of treatment irrelevance in Panel B, again at the 5 or 10 percent levels, further support the significant height-for-age effects. We interpret the Panel B results as quite strong, as the power of the height-for-age findings are enough to outweigh the lack of statistical significance in symptom and ADL outcomes in Table 3.

Overall, these adjusted inferences and joint tests strengthen the credibility of the statistical significance and precision of the height-for-age effects reported in Table 2. We thank a helpful reviewer for suggesting their inclusion.

Appendix B. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.worlddev.2020.105332.

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